MULTIPLE CHOICE QUESTIONS

QN	CORRECT ANSWER	QN	CORRECT ANSWER
1	**B**	21	**D**
2	**B**	22	**A**
3	**C**	23	**D**
4	**D**	24	**D**
5	**C**	25	**C**
6	**D**	26	**C**
7	**A**	27	**C**
8	**B**	28	**B**
9	**B**	29	**D**
10	**B**	30	**C**
11	**B**	31	**B**
12	**A**	32	**A**
13	**A**	33	**A**
14	**C**	34	**C**
15	**C**	35	**D**
16	**D**	36	**D**
17	**A**	37	**D**
18	**D**	38	**B**
19	**A**	39	**D**
20	**A**	40	**C**

STRUCTURED QUESTIONS

QUESTION 1

(a) (i) State **two** secondary structures commonly found in proteins. [1]

α-helix and β-pleated sheet ;;

(ii) Compare the secondary structures stated in **(a)(i)**. [2]

- Structures are formed as a result of regular/ repetitive, coiling / folding, of the polypeptide chain ;;
- The α-helix and β-pleated sheet are stabilised by intrachain hydrogen bonds between carbonyl (C=O) and amine (–NH) polypeptide backbone ;;
- The α-helix takes the form of an extended spiral spring/ coiled conformation whereas the β-pleated sheet has an extended sheet-like / pleated / zigzag conformation ;;

(b) Describe how the catalytic and binding amino acid residues of DNA polymerase located at different positions are brought close together in the active site. [3]

- Polypeptide chain is first folded and coiled into its secondary structures α-helix and β-pleated sheet ;;
- into specific 3D conformation of active site formed by catalytic and binding amino acid residues;;
- involving various interactions between the R groups of the structural amino acid residues *via* hydrophobic interactions, ionic bonds, hydrogen bonds and disulfide bonds/ covalent bonds (any two);;

(c) Explain the significance of the pH 8.4 buffer in PCR. [2]

- optimum pH ;;
- the rate of reaction is at a maximum/ highest amount of products formed per unit time ;;

QUESTION 2

(a)(i) state the duration of metaphase in the cell. [1]

18min

(ii) complete line Y on the graph. [1]

Horizontal until 18 minutes, then decreases as straight line to 0 μm at 28 minutes

(iii) explain your answer in **(a)(ii)**. [3]

- Chromosomes align singly at the metaphase plate during metaphase of mitosis.
- Sister chromatids separate at the centromere to become daughter chromosomes and migrate towards the opposite poles in anaphase.
- Each chromatid / daughter chromosome did not move / remain at the pole in telophase.

**quoting of data is necessary to support the answers above.*

(b) Suggest and explain the effect of eribulin on the behavior of chromosomes in mitosis. [3]

- Kinetochore microtubules cannot attach to the kinetochores at the centromeres of the chromosomes.
- Cells cannot progress through metaphase, so that chromosomes cannot align singly at the metaphase plate.
- Sister chromatids could not separate / remain attached in anaphase.

QUESTION 3

(a) Explain the significance of p53 protein in the regulation of Mdm2 gene expression. [3]

- As an activator, p53, binds to the enhancer;;
- May interact with mediator proteins;
- To improve recruitment of general transcription factors and RNA polymerase to the promoter;
- to form a stable transcription initiation complex (TIC) at the promoter;
- Ref. to upregulation of transcription of *Mdm 2* gene/ increase rate of transcription of *Mdm2* gene;

(b) Account for the high levels of p53 protein when DNA damage is detected in the cell. [4]

- During DNA damage, p53 protein is activated by phosphorylation of p53 at Ser15, Thr18 or Ser20 amino acid residues;;
- resulting in the 3D conformation of p53 protein to be no longer complementary to the 3D conformation of Mdm2 protein;;
- Hence p53 cannot bind to Mdm2 protein and p53/Mdm2 complex is not formed;;
- Thus, p53 is not degraded via ubiquitin system/ avoiding ubiquitin-mediated degradation to increase levels of p53 protein;;

(c) Suggest the need for ubiquitin-mediated degradation of p53 to occur. [1]

- To remove unwanted p53 protein/ maintain low levels of p53 when no DNA damage is detected;;

(d) Homozygous deletion of *Mdm2* gene in mouse germline cells results in lethality at the blastocyst stage, due to inappropriate apoptosis.
With reference to Fig. 3.2a and Fig. 3.2b, suggest how inappropriate apoptosis occurs. [3]

- With the homozygous deletion of the Mdm2 gene, no functional Mdm2 protein can be produced;;
- In normal cells without DNA damage, p53 has no Mdm2 protein to bind to/ p53/Mdm2 protein complex not formed;
- Resulting in no ubiquitin-mediated degradation of p53/Mdm2 complex/ p53 is not degraded via the ubiquitin system;
- Hence, leading to high levels of unbound p53 which activates apoptosis in cells;;

- despite having no DNA damage;

<u>QUESTION 4</u>

(a) (i) State what is meant by retrovirus. [2]

- Retroviruses are viruses with two identical copies of single stranded RNA;;
- and two molecules of reverse transcriptase ;;

(ii) Compare the <u>entry process</u> between the HIV and influenza virus. [3]

- Adsorption/ attachment of both viruses are by binding to specific cell surface receptors. ;;
- Glycoprotein gp120 on the surface of the HIV binds to CD4, a cell-surface receptor found on white blood cells/ T helper cells / macrophages of the host immune system ;
- Haemagglutinin on the influenza viral membrane binds to sialic acid-containing receptors on the host cell membrane ;
- The influenza virus enters the host cell by receptor-mediated endocytosis;
- The HIV envelope does not enter via receptor-mediated endocytosis. Instead, the HIV envelope fuses with the host cell membrane;
- Upon entry, the influenza virus forms an endosome / endocytic vesicle ;
- whereas the HIV virus does not form an endosome/ endocytic vesicle, the HIV releases the viral contents into the host cell cytoplasm.;

(b) Upon completion of the entry process, describe how the genome of HIV is inherited. [3]

- RNA is reverse transcribed to complementary DNA strand by the enzyme reverse transcriptase ;;
- The enzyme integrase catalyses the integration of the viral DNA into the host chromosome.;;
- The provirus genome is also replicated along with the host cell genome and all daughter cells inherit the HIV genome/ AW ;

QUESTION 5

(a) Explain the type of gene interaction observed in this context. [3]

- The type of gene interaction is (duplicate dominant) epistasis;;.
- Allele A is epistatic to alleles B and b , allele B is epistatic to alleles A and a;;
- The presence of at least 1 dominant allele of either gene A or B causes the conversion of precursor X to product Z hence, allowing the formation of heart-shaped fruits;;.
- If the genotypes at both loci are homozygous, the conversion of precursor X to product Z does not occur. ;;

(c) Using the symbols **A**, **a**, **B** and **b**, draw a genetic diagram to explain the result of the F2 generation in the space provided. [5]

Let A be the dominant allele for the production of heart-shaped fruit
Let a be the recessive allele for the production of narrow fruit;
Let B be the dominant allele for the production of heart-shaped fruit
Let b be the recessive allele for the production of narrow fruit;

Parental phenotype	Heart shaped fruit		Narrow fruit
Parental genotype	AABB	x	aabb ;
Parental gametes	AB		ab ;
F1 genotype		AaBb ;	
F1 phenotype		Heart-shaped fruit	
F1 self-cross	AaBb	x	AaBb
Parental gametes	AB Ab aB ab		AB Ab aB ab ;

Random fertilization

Male gametes	Female gametes: AB	Ab	aB	ab
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	AAbb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	aabb

;;

F2 genotypic ratio	9 A_B_	:	3 A_bb:	:	3 aaB_	:	1 aabb ;
F2 phenotypic ratio	15 Heart-shaped fruit					:	1 Narrow fruit
F2 observed no	2251					:	150 ;

(d) Describe how you could identify *Capsella* sp. plants with heart-shaped fruits, which are homozygous dominant in at least one gene locus. [2]

- Perform a testcross;;
- If the plants with heart-shaped fruits are homozygous dominant in at least one gene locus, all offspring produced will only have heart-shaped fruits;;.

<u>QUESTION 6</u>

(a) **(i)** State precisely where RuBP and GP are produced in the chloroplast. [1]

stroma

(ii) Explain why the concentration of RuBP changed between 200 and 275 seconds. [2]

- lower CO_2 concentration
- concentration of RuBP increases from 1 au to 1.6 au
- less CO_2 fixed by RuBP

(b) Suggest how the decrease in the concentration of GP leads to a decrease in harvest for commercial suppliers of *Chlorella*. [2]

- less triose phosphate / glyceraldehyde-3-phosphate will be produced
- so less conversion to carbohydrates / lipids / amino acids / proteins

(c) In the light dependent stage, illumination of chloroplasts is important for maintaining the high pH in the stroma. Explain how the illumination of chloroplasts maintains the high pH in the stroma. [3]

- excited electrons leave, special chlorophyll *a*
- electron passed down the electron transport chain
- protons pumped into thylakoid lumen from the stroma
- protons present from photolysis of water

(d) The endosymbiotic theory postulates that the chloroplasts of *Chlorella* evolved from bacteria living within an eukaryotic host cell. Suggest **one** structural similarity between the chloroplasts and the bacteria that supports this theory. [1]

- ref. to chloroplasts having a double membrane / 70S ribosomes

QUESTION 7

(a) (i) Draw an arrow in the box provided on Fig. 7.1 to indicate the direction in which one nerve impulse is being conducted and explain your answer. [1]

Arrow pointing to the left ;;

(ii) Explain you answer in (a)(i). [2]

- Hyperpolarisation occurs during the refractory period;;
- Hence the nerve impulse can only travel towards the left where depolarisation occurs. ;;

(b) Suggest why transmission of nerve impulses along a myelinated neurone uses less energy in the form of ATP than transmission along an unmyelinated neurone. [2]

- For myelinated neurone, charges can only be leaked through the nodes of Ranvier, compared to myelinated neurone, where charges can be leaked throughout the neurone ;;
- Hence, less energy in the form of ATP is required by active transport pumps, Na^+/K^+ pumps to restore the resting membrane potential ;;

(c) Account for the changes in the concentration of sodium ion at region X upon the stimulation of the neurone. [4]

- Upon the stimulation of the neurone above threshold potential, depolarisation occurs at the axon hillock and voltage-gated Na^+ channels open ;
- Na^+ influx / Na^+ enters the neurone ;;
- resulting in sudden increase/ spike in concentration of sodium inside the neurone ;
- subsequently, repolarisation occurs and voltage-gated Na^+ channel close / membrane becomes less permeable to Na^+ ;
- Na^+/K^+ pumps will restore concentration of sodium in the neurone ;;
- leading to gradual decrease in concentration of Na^+ until resting membrane potential is achieved ;

(d) Suggest and explain why the concentration of sodium ions did remain constant at region Y. [3]

- Proton gradient cannot be maintained ;;
- ATP cannot be generated via chemiosmosis ;;
- Ref. to sodium ions entering the neurone but cannot be transported out of the cell;;

QUESTION 8

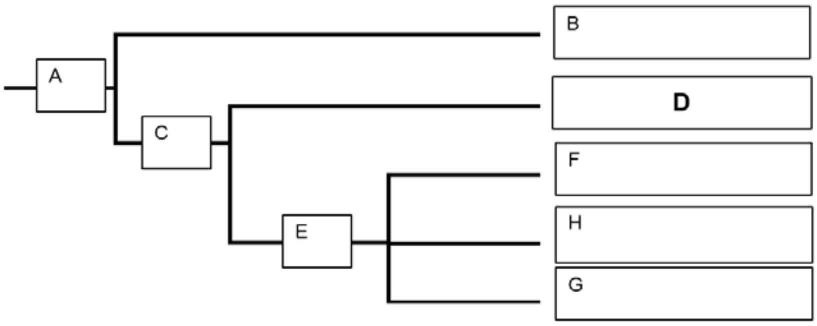
(a) Explain how the anatomical similarities among the whales, pigs and hippopotamuses support Darwin's theory of natural selection. [3]

- Limb and inner ear bones were both present in the ancestor / Cetartiodactyls;
- The presence of homologous structures / anatomical homologies between the pigs, hippopotamus and the whales signify shared / common ancestry / evolutionary relatedness;
- These animals will survive to sexual maturity / survival of the fittest / differential reproduction to pass on the favourable genes / alleles to their offspring;.
- leading to genetic variation that brings about differences in limb and inner ear bone thickness;.
- Hence showing decent with modification;

(b)(i) Explain why camels and whales are of different species according to the phylogenetic species concept. [3]

- In the phylogenetic species concept, a species is defined as the smallest group of individuals that share a common ancestor;;
- Camels and whales do not share a most recent common ancestor;
- The camels are the earliest to diverge from the ancestor of cetartiodactyls ;
- as they do not possess sequences of SINE 1 in their genome / ref made to shared derived characteristics with quotation;
- Because they possess SINE1 / SINE 1 & 2 in their genome / ref made to shared derived characteristics with quotation;

(ii) A phylogenetic tree is constructed based on the results of SINE analysis in Fig 8.1. Fill in the blanks at the branch points and end points of the phylogenetic tree with the corresponding letters, **A** to **H** as represented in Fig. 8.1. [2]



(c) Briefly describe the neutral theory of molecular evolution and suggest how it may be used to study the subgroups of Cetartiodactyls. [4]

- The neutral theory states that vast majority of evolutionary change at the molecular level are caused by random genetic drift of selectively neutral mutations;;
- which do not affect the phenotype / fitness of the organism / not subjected to natural selection / has neither selective advantage or disadvantage.;;
- Since SINE sequences are non-coding, mutations in them are not affected by natural selection;
- The greater the difference in number of mutations;
- the longer time of divergence.;;

(d) Outline how the organization of the genome allows for DNA packing in a non-dividing cell. [2]

- DNA molecule is wound around histone proteins to form the **nucleosome;;**
- Nucleosomes are linked by liner DNA to form **10nm fibers chromatin**;
- nucleosomes are further packed into **30-nm fibres / solenoids;**

FREE RESPONSE QUESTION

QUESTION 9

(a) Describe how the molecular structure of cellulose is related to its support function. [6]

- Alternate inverted β-glucose units linked by β(1,4) glycosidic bonds allow cellulose to form long, unbranched and straight chains;;
- Allow formation of linear chains of polysaccharides that can be packed tightly;;
- Many chains run parallel to each other and their hydroxyl group (OH) project outwards from each chain;;
- Extensive hydrogen bonds form between parallel chains/ Extensive hydrogen bonds form between the protruding OH groups of neighbouring chains;;
- Cross-linked cellulose chains associate in groups to form microfibrils;;
- Microfibrils associate with other, non-cellulose polysaccharides, and are arranged in large bundles to form macrofibrils;;
- Allows formation of a large molecule, resulting in an insoluble material that can be used as structural support;;

(b) Outline the basis of the selective permeability of the cell membrane with reference to phospholipids, cholesterol and proteins. [8]

Phospholipids

- cell membrane is a phospholipid bilayer ;
- it acts as a hydrophobic barrier / ref. to hydrophobic core ;;
- which prevents the diffusion of hydrophilic solutes / polar molecules, charged ions across it ;;
- weak hydrophobic interactions exist between phospholipids / ref. to lateral movement of phospholipids ;
- for small, non-polar, hydrophobic solutes to diffuse across ;;

Cholesterol

- presence of cholesterol in the membrane decreases the permeability of membrane ;;
- it does so by filling in spaces between hydrocarbon chains of phospholipids / plugging transient gaps ;;
- cholesterol regulates membrane fluidity / ref. to higher temperature, cholesterol decreases membrane fluidity / lower temperature, cholesterol increases membrane fluidity ;;

Proteins

- hydrophilic solutes / polar molecules, charged ions can be transported across the membrane ;;
- through transport proteins ;;
- ref. to solute binding and change its 3D conformation ;
- via facilitated diffusion / ref. to down a concentration gradient ;;
- via active transport / ref. to against a concentration gradient ;;
- presence of channel proteins ;;
- ref. to water-filled central pore / hydrophilic channel ;
- via facilitated diffusion / ref. to down a concentration gradient ;;

(c) Explain why animal cells mainly store lipids instead of carbohydrates. [6]

- Triglycerides are a good thermal insulator and hence a layer of fat beneath the skin (subcutaneous fat) insulates the body. This subcutaneous layer is especially thick in whales, seals and most other marine animals living in cold climates and is known as blubber;;
- Upon oxidation, triglycerides release a larger amount of energy per unit / ref. to one gram of fat releases more than twice as much energy (38 kJ / g) as a gram of carbohydrates (17 kJ / g);;
- Being highly reduced molecules, triglycerides release <u>more</u> metabolic water when they are oxidised during cellular respiration compared to carbohydrates, which is extremely important to desert animals like camels;;
- Triglyceride can slide under pressure hence adipose tissue (which contains fats) around the vital organs helps to cushion and protect vital organs against physical impacts;;
- Being less dense than water, fats aid buoyancy of aquatic animals ;;

<u>QUESTION 10</u>

(a) Describe how the molecular structure of the GPCR is suited for its role in glucagon- mediated cell signaling. [8]

- The GPCR is a transmembrane protein comprising an extracellular domain, a transmembrane region and a intracellular domain ;;
- It relays an extracellular signal to a cellular response ;;
- The extracellular domain of GPCR has a ligand binding site that is complementary to the 3D conformation of glucagon;;
- This allows glucagon to bind to its specific GPCR on the target cells in signal reception ;;
- GPCR consists of a single polypeptide chain that has seven α- helices spanning the membrane;;
- The hydrophobic interactions between the non-polar R-groups with the fatty acid tails of phospholipid bilayer allows the GPCR to be embedded in the membrane ;;
- The intracellular domain of GPCR is complementary to the 3D conformation of a G protein;;
- The activated GPCR binds and activates the G protein for signal transduction ;;

(b) Outline the concept of negative feedback in controlling glucagon levels in the body. [6]

- When blood glucose levels falls below set point of 90mg/100ml, the receptor, alpha cells in the islets of Langerhans of the pancreas, detects the change / deviation;;
- This information is integrated at the integrating centre, to initiate a response in the form of signals to act upon target / effector cells;;
- the alpha cells of the islets of Langerhans of the pancreas are then stimulated to release more glucagon;;
- Glucagon binds to GPCR receptors to activate the GPCR receptors;;
- Cellular responses are stimulated when adenylyl cyclase is activated to produce cAMP to activate ;;
- When blood glucose levels are detected by the receptor to return to the set point, negative feedback mechanism is activated to prevent further increase in blood glucose beyond setpoint;;.
- There is decreased stimulation of alpha cells resulting in decreased release of glucagon;;

(c) Explain how signal amplification is illustrated by the binding of insulin to its receptor. [6]

- The binding of small number of insulin to its receptors can lead to the activation of many glycogen synthase / AW ;; to facilitate glucose uptake and storage.
- Cytoplasmic relay proteins specific to the insulin receptor binds to a specific phosphorylated tyrosine on the receptor and undergo conformational changes;.
- These bound relay proteins becomes activated;;.
- The activated relay proteins triggers a signal transduction pathway;;
- whereby activated relay proteins (Accept: protein kinases) activate other protein kinases in phosphorylation cascade;.
- Many activated downstream protein also leads to the activation of glycogen synthase which facilitates glycogenesis / AW;.